Kommutator portlari xavfsizligi

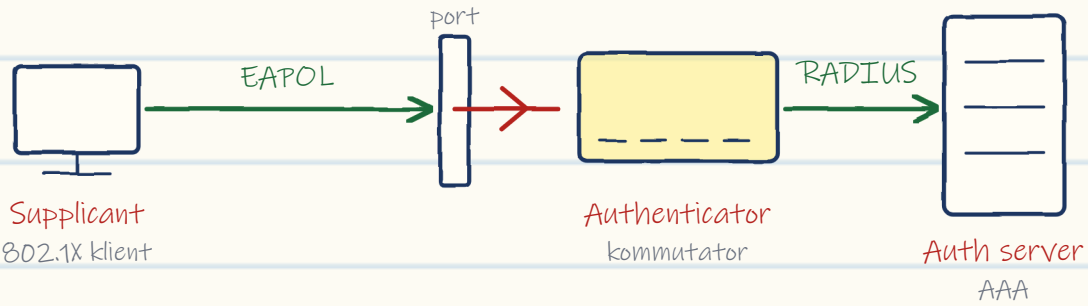
Port Security for Switches

Asosiy atamalar
- 802.1X — port asosida foydalanuvchini tekshirish protokoli
- Supplicant — kirmoqchi bo'lgan mijoz (PC) dasturi
- Authenticator — tekshiruvni o'tkazuvchi kommutator
- RADIUS server — parol va huquqlarni saqlovchi server
- Port izolyatsiyasi — portlar o'zaro gaplashmaydi, uplink ochiq

1. 802.1X — portda kirish nazorati

IEEE 802.1X — port asosida kirishni nazorat qilish (port-based access control), 2001-y.

- 802.11 WLAN uchun ishlangan — endi simli LAN'da ham
- tekshiruvdan o'tmagan klient uchun port yopiq



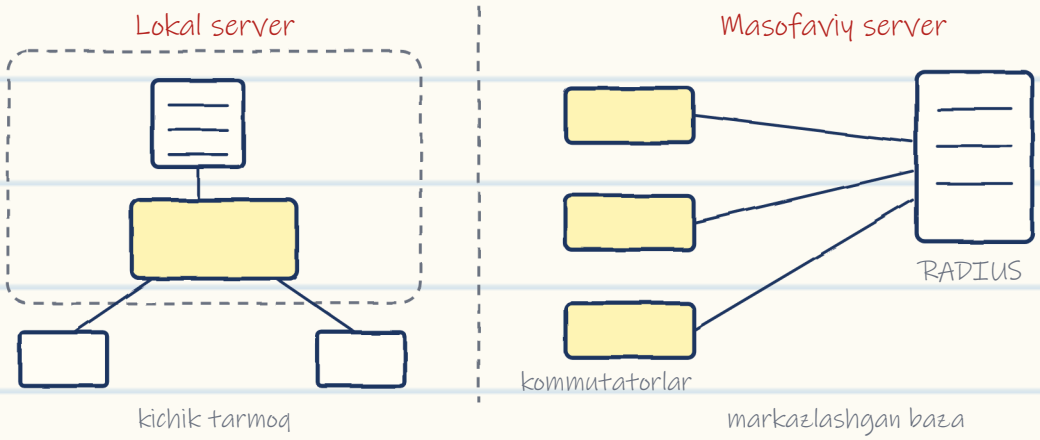
o'tmasa — kabeli uzilgandek

1-rasm. Uchta rol: klient — kommutator — server

- Supplicant — 802.1X klienti bor qurilma, EAPOL yuboradi
- Authenticator — kommutator: portni ochadi yoki yopadi
- Auth server — AAA: autentifikatsiya · avtorizatsiya · hisob

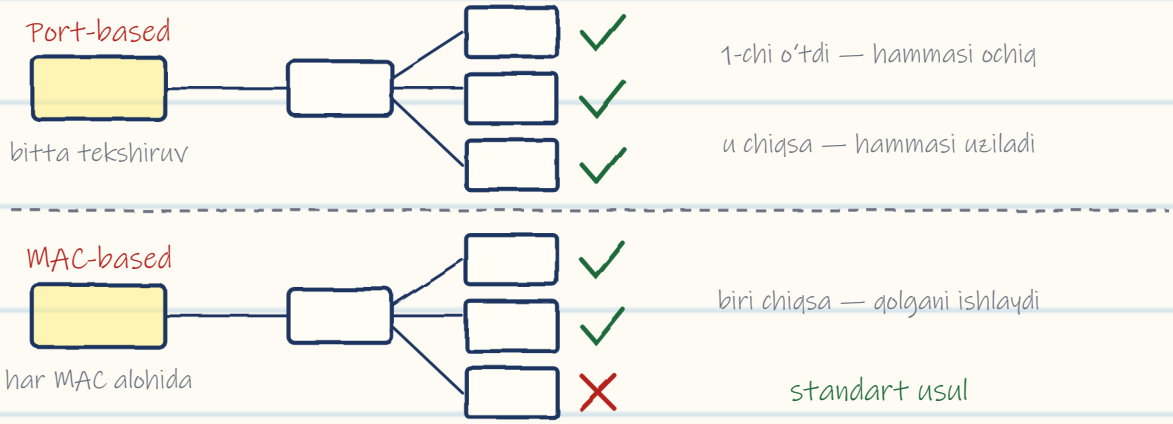
> Yo'nalish: klient ↔ kommutator — EAPOL, kommutator ↔ server — RADIUS.

2. Server: lokal yoki masofaviy



- lokal — kommutator ichida; oz foydalanuvchi, ma'lumot tarqoq
- masofaviy — alohida server; katta tarmoq, bitta baza

3. Kirish nazorati usullari

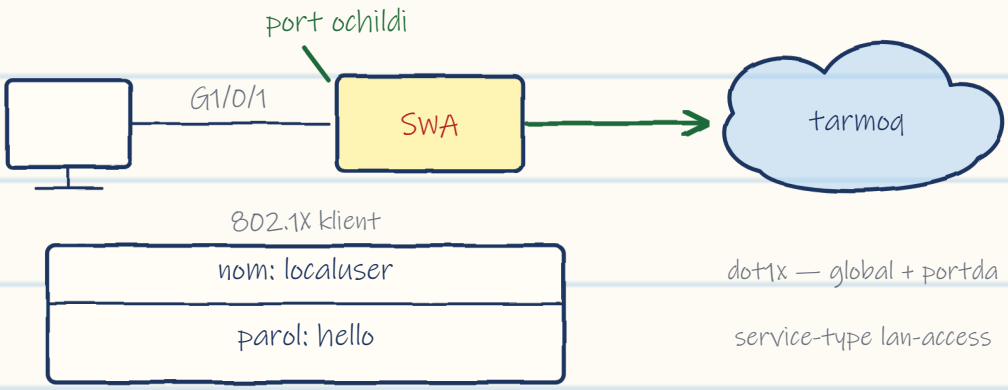


	Port-based	MAC-based
Tekshiruv	portda bir marta	har foydalanuvchi
Biri chiqsa	hammasi uziladi	faqat o'zi

4. 802.1X ni sozlash

- 1) tizim ko'rinishi: dot1x
- 2) port ko'rinishi: dot1x
- 3) local-user nomi class network · service-type lan-access · password simple parol

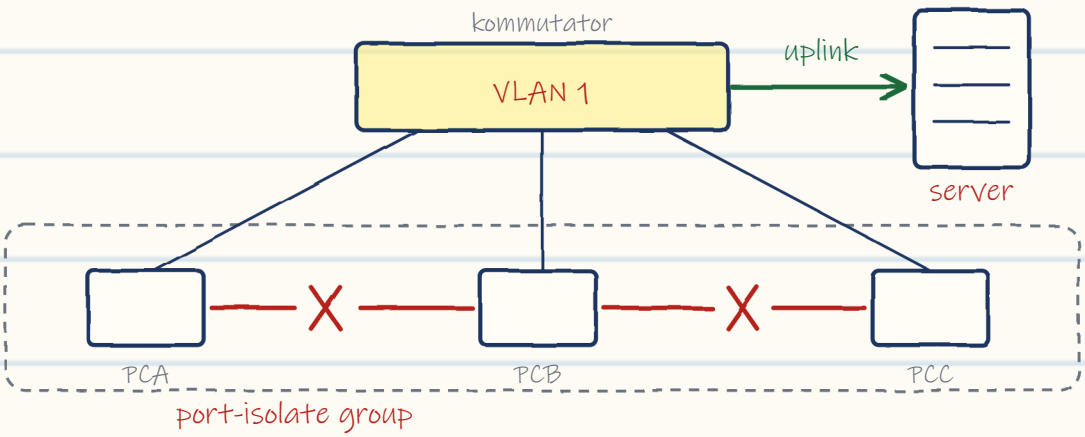
> Qoida: global va portda — ikkalasida ham yoqilmasa, 802.1X ishlamaydi.



2-rasm. PC → G1/0/1 · lokal 802.1X tekshiruvi

5. Port izolyatsiyasi

Bitta VLAN ichidagi portlarni ajratadi — VLAN resursi tejaladi.

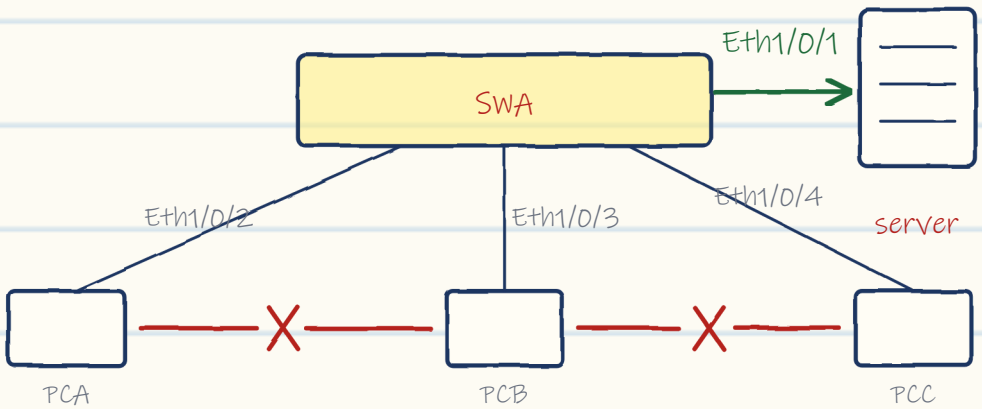


3-rasm. Guruh a'zolari o'zaro yopiq, uplink orqali ochiq

6. Izolyatsiyani sozlash

- tizim: port-isolate group 1 — guruh yaratish
- port: port-isolate enable group 1 — oddiy a'zo
- port: port-isolate uplink-port — yuqoriga chiqish porti

> Eslatma: 1-guruhni tizim o'zi yaratadi.



2-guruh: Eth1/0/2 – 1/0/4

4-rasm. Uchta PC o'zaro yopiq, serverga yo'l ochiq

Xulosa

- 802.1X — portda kirish nazorati, IEEE, 2001-y.
- Rollar: supplicant · authenticator · auth server
- Server lokal (kichik) yoki masofaviy (katta tarmoq)
- Usul: port-based yoki MAC-based (standart)
- Port izolyatsiyasi — bir VLAN ichida portlarni ajratadi

Tekshirish

1. 802.1X qaysi tashkilot standarti? — IEEE, 2001-y.
2. Uchta rol? — supplicant, authenticator, server
3. Portdagi standart usul? — MAC-based
4. Uplink porti buyrug'i? — port-isolate uplink-port